

Cyber Security CTF Machine Development Report

Title

Cyber Security CTF Machine Development Assignment

Submitted By:

Neha Singh T (R25DE099)

Course: Cyber Security

Operating System: Kali Linux

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1. Objective

The objective of this assignment was to design and deploy a vulnerable Capture The Flag (CTF) machine on a Linux environment. The machine simulates a real-world penetration testing target where participants perform reconnaissance, enumeration, web exploitation, SSH access, privilege escalation, and steganography to capture flags. The machine is intentionally configured with hidden clues that require logical analysis instead of direct disclosure of credentials.

2. Machine Specifications

Specification	Details
Virtualization Software	VirtualBox
Operating System	Kali Linux
Web Server	Apache2
SSH Service	OpenSSH
Hostname	Neha Singh
Firewall	UFW
Browser Tested	Firefox

3. Operating System Used

The operating system used for developing the CTF machine is Kali Linux.

Kali Linux provides a stable environment for hosting vulnerable machines while supporting security testing tools required for penetration testing exercises.

4. Network Configuration

The machine was configured using a bridged/network adapter allowing communication with other systems on the same network.

Accessible ports:

Port 22 (SSH)

Port 80 (HTTP)

All unnecessary ports were blocked using the firewall.

5. User Accounts Created

Three separate Linux user accounts were created.

Each user has:

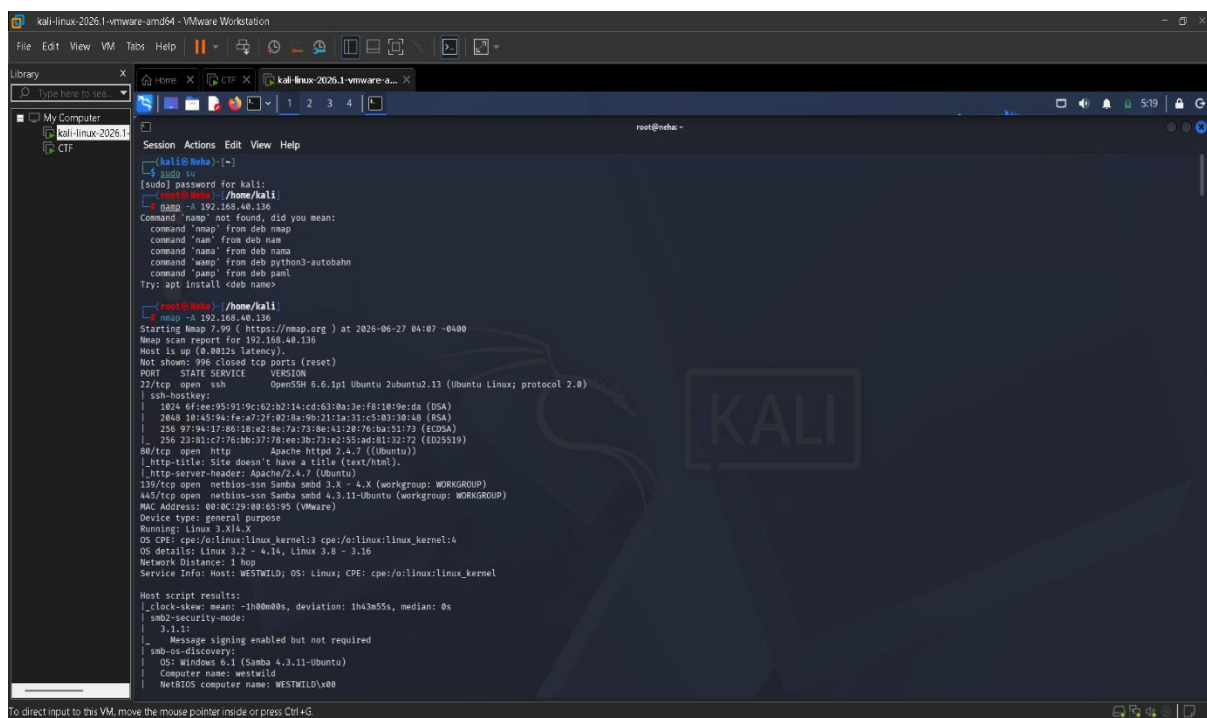
Individual home directory

Login shell

Hidden credentials

For security purposes, usernames and passwords are not disclosed within this report.

Instead, they are hidden inside the website and require enumeration by participants.



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1
CTF
Session Actions Edit View Help
root@kali: ~
[~]
[~] sudo su
[sudo] password for kali:
[~] root@kali: /home/kali
[~] nmap -A 192.168.40.136
Command 'nmap' not found, did you mean:
command 'nmap' from deb nmap
command 'nan' from deb nan
command 'nana' from deb nana
command 'wamp' from deb python3-autobahn
command 'paep' from deb paep
Try: apt install cdeb nmap
[~] root@kali: /home/kali
[~] nmap -A 192.168.40.136
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 04:07 -0400
Nmap scan report for 192.168.40.136
Host is up (0.0025s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_ 1024 ff:ee:95:91:9c:62:b2:14:cd:63:8a:3e:f8:10:9e:da (DSA)
|_ 2048 8f:45:94:fe:a7:2f:f2:8e:09:21:5a:31:c5:03:30:46 (RSA)
|_ 256 07:9a:17:06:18:e2:8e:7a:73:9e:42:28:76:ba:51:73 (ECDSA)
|_ 256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: site doesn't have a title (text/html).
|_ http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   open  netbios-ssn  Samba smb 3.x - 4.x (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smb 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 08:0C:29:00:65:95 (VMware)
Device type: general purpose
Running: Linux 3.14.x
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel
Host script results:
|_ clock-skew: mean: -1h40m0s, deviation: 1h43m55s, median: 0s
|_ smb2-security-mode:
|_ 3.1.1:
|_ Message signing enabled but not required
|_ smb-os-discovery:
|_ OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|_ Computer name: westwild
|_ NetBIOS computer name: WESTWILD\*000
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

6. Services Configured

The following services were configured.

Apache Web Server

Runs on Port 80

Hosts personal portfolio

Contains hidden clues

SSH

Runs on Port 22

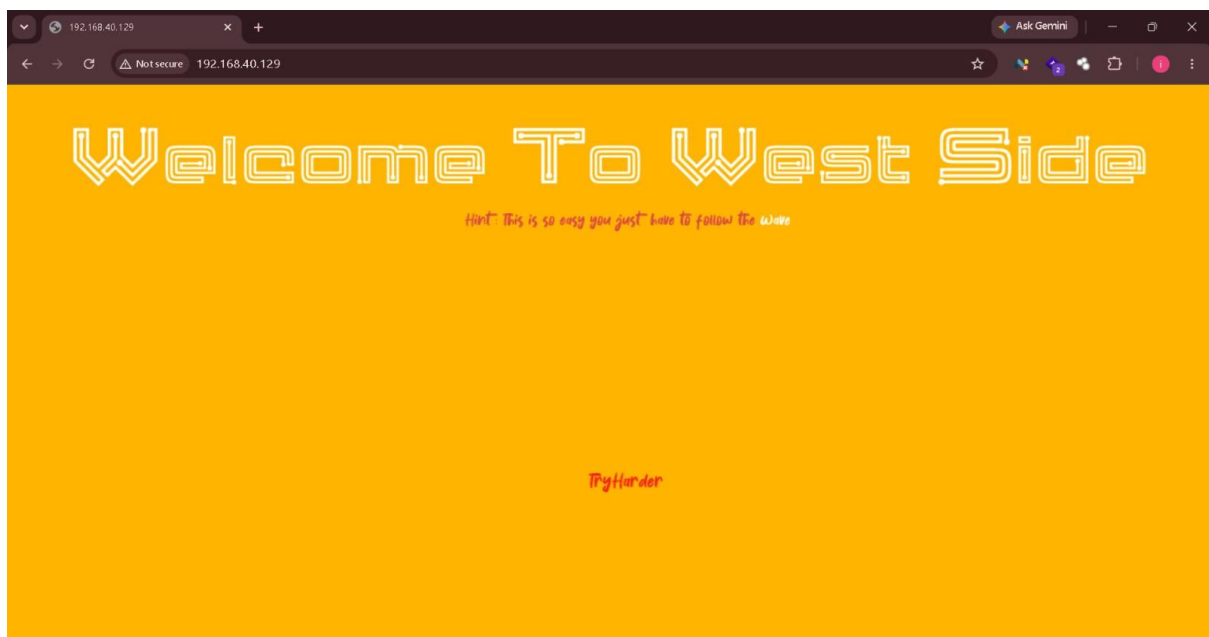
Used after discovering credentials

Firewall

Configured using UFW allowing only:

SSH

HTTP



7. Website Design Overview

A personal portfolio website was developed containing:

Professional profile

Skills

Certifications

Projects

Contact information

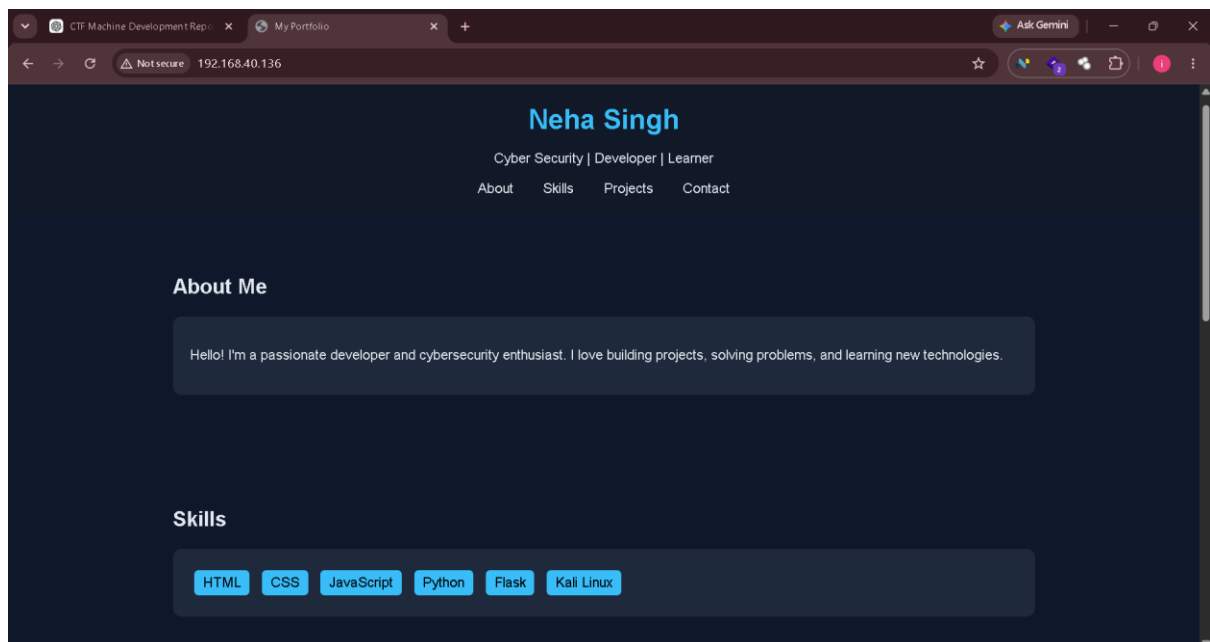
The website also contains intentionally hidden information for participants to discover.

Website technologies:

HTML

CSS

JavaScript



8. Hidden Credentials Methodology

The machine requires enumeration.

Hidden information was distributed using several techniques including:

HTML comments

JavaScript source files

CSS comments

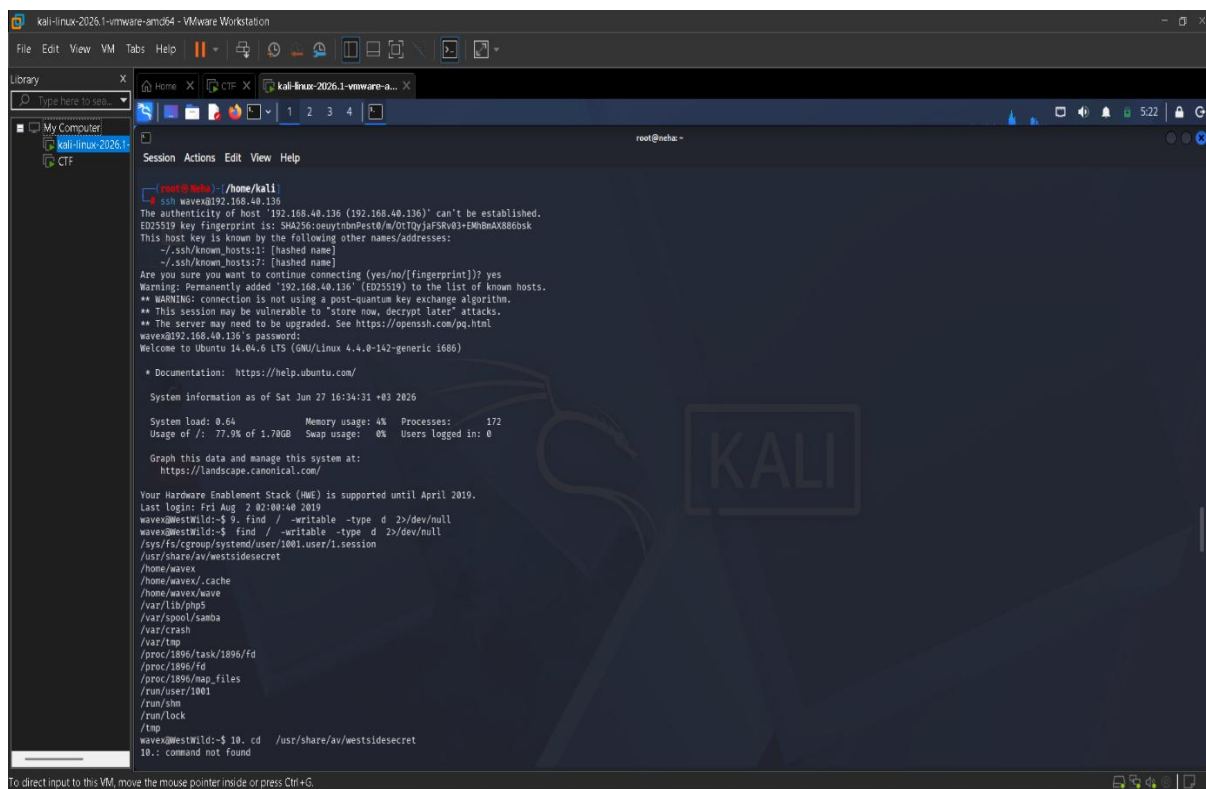
robots.txt

Hidden directories

Encoded strings (Base64)

Source code inspection

These techniques encourage participants to perform realistic web enumeration.



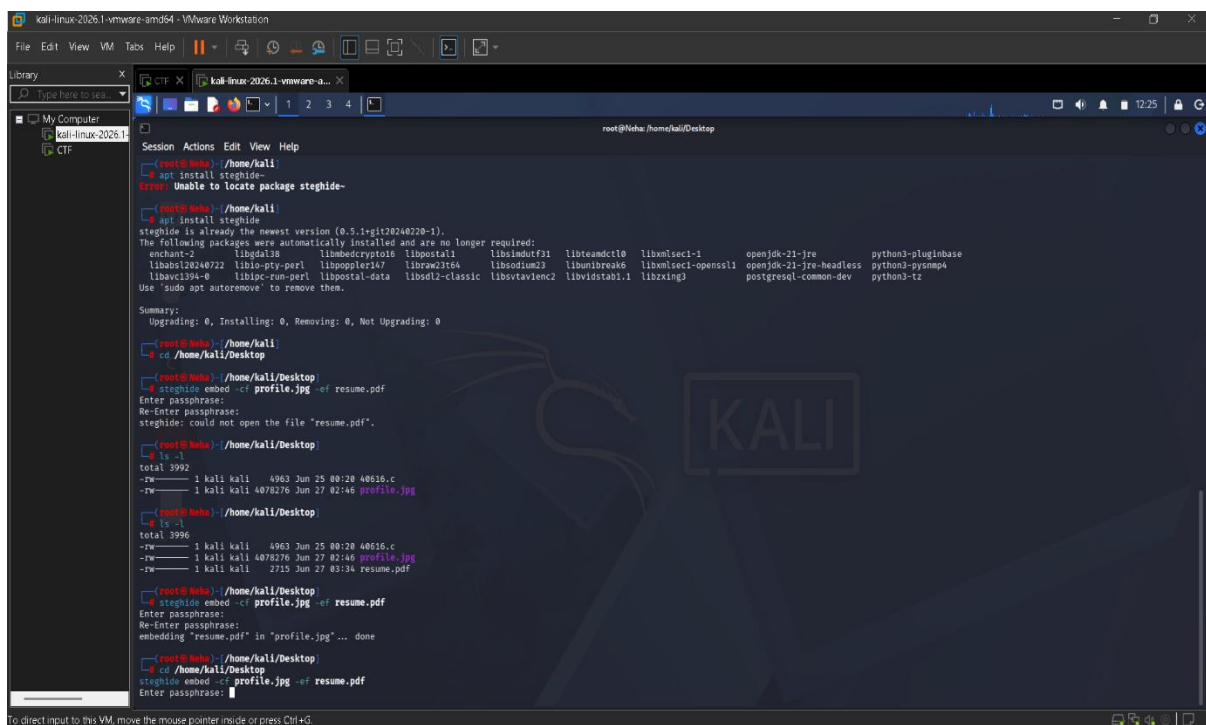
```
root@kali: ~  
root@kali: ~/kall  
root@kali:~# ssh wavex@192.168.40.136  
The authenticity of host '192.168.40.136 (192.168.40.136)' can't be established.  
ED25519 key fingerprint is: SHA256:qeuYtnbnPest0/a/OtQyjaFSRv03+EMhbnAX886bdk  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [dashed name]  
  ~/.ssh/known_hosts:7: [dashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.40.136' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
wavex@192.168.40.136's password:  
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)  
  
* Documentation: https://help.ubuntu.com/  
  
System information as of Sat Jun 27 16:34:31 +03 2026  
  
System load: 0.64      Memory usage: 4%      Processes:    172  
Usage of /: 77.9% of 1.7GB  Swap usage:   0%      Users logged in: 0  
  
Graph this data and manage this system at:  
https://landscape.canonical.com/  
  
Your Hardware Enablement Stack (HWE) is supported until April 2019.  
Last login: Fri Aug 2 02:18:40 2019  
wavex@westwild:~$ find / -writable -type d 2>/dev/null  
wavex@westwild:~$ find / -writable -type d 2>/dev/null  
/sys/fs/cgroup/systemd/user/1001/user/1/session  
/usr/share/av/westsidecrypt  
/home/wavex  
/home/wavex/.cache  
/home/wavex/wave  
/var/lib/php5  
/var/spool/samba  
/var/crash  
/var/tmp  
/proc/1896/task/1896/fd  
/proc/1896/fd  
/proc/1896/map_files  
/run/user/1001  
/run/shm  
/run/lock  
/tmp  
wavex@westwild:~$ cd /usr/share/av/westsidecrypt  
10.: command not found
```

9. Steganography Implementation

A file named profile.jpg was placed on the desktop.

A professionally formatted resume was embedded into the image using steganography.

Participants must identify the hidden file and extract it using appropriate forensic tools.



```

kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1
CTF
root@kali: /home/kali
# apt install steghide
Error: Unable to locate package steghide-
# apt install steghide
steghide is already the newest version (0.5.1git120240220-1).
The following packages were automatically installed and are no longer required:
  enchant-2 libgdal38 libimbedcrypto6 libpostal1 libsimdutf31 libteardict0 libxmlsec1-1 openjdk-21-jre python3-pluginbase
  libnsl120240722 libio-pty-perl libpoppler247 librew23164 libssodium2 libunibreak6 libxmlsec1-openssl1 openjdk-21-jre-headless python3-pysnmp4
  libavc1394-0 libipc-run-perl libpostal-data libssdl2-classic libsvtavcodec2 libvidstab1.1 libzxing3 postgresql-common-dev python3-tz
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
# cd /home/kali/Desktop
# steghide embed -c profile.jpg -ef resume.pdf
Enter passphrase:
Re-Enter passphrase:
steghide: could not open the file "resume.pdf".
# ls -l
total 3992
-rw-r--r-- 1 kali kali 4963 Jun 25 00:28 40616.c
-rw-r--r-- 1 kali kali 4078276 Jun 27 02:46 profile.jpg
# ls -l
total 3996
-rw-r--r-- 1 kali kali 4963 Jun 25 00:28 40616.c
-rw-r--r-- 1 kali kali 4078276 Jun 27 02:46 profile.jpg
-rw-r--r-- 1 kali kali 2715 Jun 27 03:34 resume.pdf
# steghide embed -c profile.jpg -ef resume.pdf
Enter passphrase:
Re-Enter passphrase:
embedding "resume.pdf" in "profile.jpg"... done
# cd /home/kali/Desktop
# steghide embed -c profile.jpg -ef resume.pdf
Enter passphrase:

```

Example extraction:

steghide extract-sf profile.jpg

10. Flag Locations

Two flags were created.

User flag

Root flag

```

kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1
CTF
Session Actions Edit View Help
root@neha: /var/www/html
exit
$ ls
bin boot dev etc home initrd.img lib lost-found media mnt opt proc root run shin srv sys tmp usr var vmlinuz
$ su
Password:
root@neha: /# cd /home/wavex/wave
root@neha: /home/wavex/wave# ls
FLAG1.txt message_from_avenq.txt
root@neha: /home/wavex/wave# rm
rm: missing operand
Try 'rm --help' for more information.
root@neha: /home/wavex/wave# rm message_from_avenq.txt
root@neha: /home/wavex/wave# echo "user=dev:pass=root1234" | base64
echo: user=dev:pass=root1234: command not found
root@neha: /home/wavex/wave# echo "user=dev:pass=dev123" | base64
echo: user=dev:pass=dev123: command not found
root@neha: /home/wavex/wave# echo "user=dev:pass=dev123" | base64
dXNlcjJZYXZlZC9zc2lkZXN0MjM0
root@neha: /home/wavex/wave# vim FLAG1.txt
root@neha: /home/wavex/wave# vim FLAG1.txt
root@neha: /home/wavex/wave# vim FLAG1.txt
root@neha: /home/wavex/wave# cd /usr/share/av/westsidecret
bash: cd /usr/share/av/westsidecret: No such file or directory
root@neha: /home/wavex/wave# cd /usr/share/av/westsidecret
root@neha: /usr/share/av/westsidecret# vim ififoregt.sh
root@neha: /usr/share/av/westsidecret# cd
root@neha: /# vim FLAG2.txt
root@neha: /#
:::1 ff02::1 ff02::2 ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost Westwild
root@neha: /#
:::2 ff02::1 ff02::2 ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost Westwild
root@neha: /#
:::1 ff02::1 ff02::2 ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost Westwild
root@neha: /# sudo apt update
Get:1 http://security.ubuntu.com trusty-security InRelease [56.5 kB]
Ign http://us.archive.ubuntu.com trusty InRelease
Get:2 http://us.archive.ubuntu.com trusty-updates InRelease [56.5 kB]
Get:3 http://security.ubuntu.com trusty-security/main Sources [254 kB]
Get:4 http://security.ubuntu.com trusty-security/restricted Sources [7,511 B]
Get:5 http://security.ubuntu.com trusty-security/universe Sources [154 kB]
Get:6 http://security.ubuntu.com trusty-security/multiverse Sources [4,388 B]
Get:7 http://security.ubuntu.com trusty-security/main 1286 Packages [690 kB]
Get:8 http://security.ubuntu.com trusty-security/restricted 1386 Packages [19.3 kB]
Get:9 http://security.ubuntu.com trusty-security/universe 1386 Packages [385 kB]
Get:10 http://security.ubuntu.com trusty-security/multiverse 1386 Packages [6,739 B]
Get:11 http://security.ubuntu.com trusty-security/main Translation-en [225 kB]
Get:12 http://us.archive.ubuntu.com trusty-backports InRelease [56.5 kB]

```

11. Hostname Configuration

The system hostname was changed to: Neha

13. Testing Procedure

The machine was tested using the following methodology.

Step 1

Port scanning

nmap



Step 2

Visit website



Step 3

Inspect source code



Step 4

Check robots.txt



Step 5

Enumerate hidden files



Step 6

Decode hidden strings



Step 7

Recover credentials



Step 8

SSH Login



Step 9

Capture user flag



Step 10

Privilege escalation



Step 11

Capture root flag



Step 12

Extract hidden resume from profile.jpg

14. Vulnerability Walkthrough

The attack path implemented in this machine is as follows:

1. Perform network scan to identify open services.
2. Access the hosted portfolio website.

3. Inspect HTML, CSS, JavaScript, and robots.txt for hidden clues.
4. Decode encoded hints to discover usernames and passwords.
5. Authenticate via SSH using the recovered credentials.
6. Retrieve the user flag from the user's home directory.
7. Escalate privileges to the root account.
8. Retrieve the root flag.
9. Analyze the image placed on the Desktop and extract the embedded resume using steganography.

This walkthrough demonstrates the importance of secure web development, proper credential management, and defense against information disclosure.

15. Conclusion

This project demonstrates the design and deployment of a realistic Capture The Flag machine using Kali Linux. The machine integrates web enumeration, credential discovery, SSH authentication, privilege escalation, firewall configuration, and steganography into a single practical exercise.

The assignment enhanced practical skills in Linux administration, web server configuration, security hardening, and offensive security techniques. It also provided hands-on experience in creating challenges that reflect real-world penetration testing scenarios while reinforcing the importance of identifying and mitigating common security weaknesses.

Appendix

Software Used

Kali Linux

Apache2

OpenSSH

UFW Firewall

Steghide

Nano Editor

Nmap

Firefox Browser .